

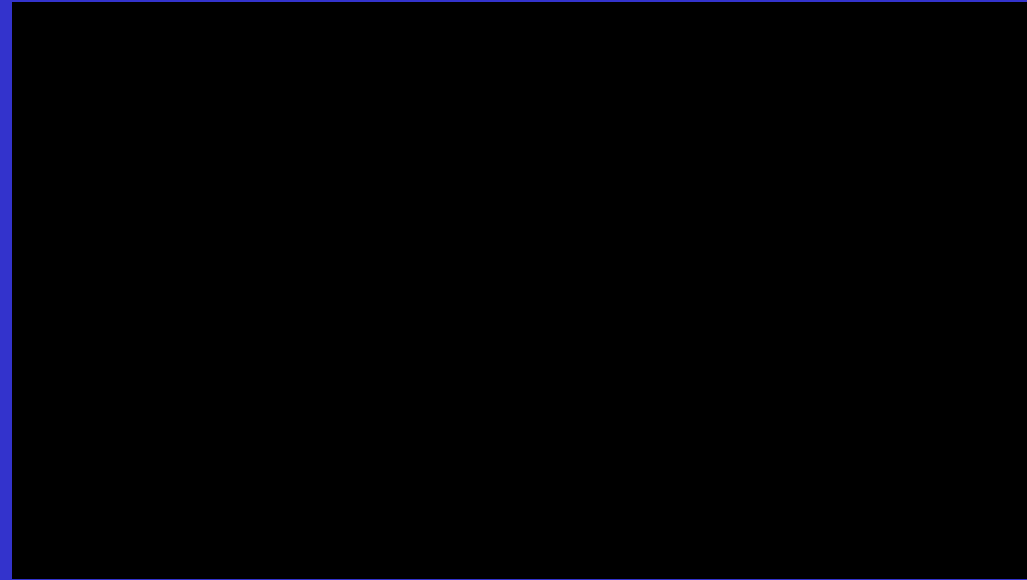
Metal Forming Technology - Elements of the Theory of Plasticity





CLASSIFICATIONS OF FORMING PROCESSES

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Metal Forming → The easiness with which metals can be formed into useful shapes such as tubes, rod, and sheets.

1. By *plastic deformation/ mechanical working processes* → volume and mass of metal are conserved; metal is displaced from one location to another.
2. By *metal removal or machining processes* → material is removed in order to get the required shape.



Mechanical working processes or Manufacturing processes:

- 1) Casting (Continuous)
- 2) Rolling, Forging, Bending
- 3) Extrusion
- 4) Shearing, Piercing etc.,



Metal forming processes [Metal is conceived from one shape to another. Metal is not removed]

Metal Joining Processes are:

- 1) Welding
- 2) Brazing
- 3) Soldering
- 4) Diffusion Bonding Type
- 5) Powder Metallurgy
- 6) Riveting, Shrink Fitting [for bearings]

Metal Cutting Processes or Machining Processes :

[Extra metal is removed]

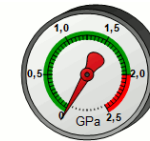
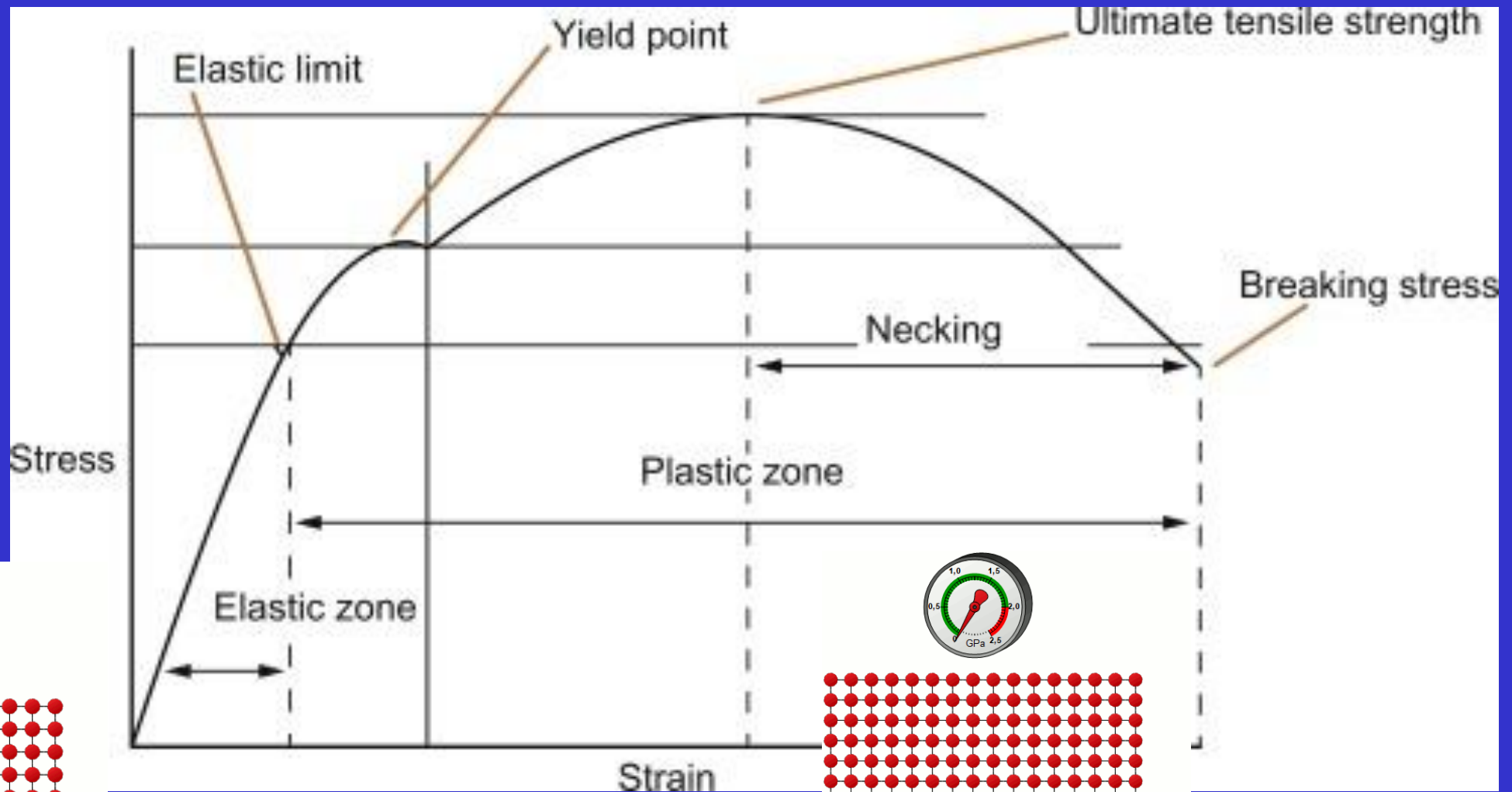
- 1) Lathe
- 2) Thread cutting
- 3) Grinding
- 4) Knurling
- 5) Drilling

Advantages of mechanical working →

Control of mechanical properties to desired level

Improvement in ductility and fracture toughness, by eliminating defects such as blowholes and porosity in a cast ingot

Surface finish and dimensional accuracy is achieved (by controlling the strain hardening during processing).



Elastic
Deformation

Plastic
Deformation

Classifications of Mechanical working Processes:

1. Based on the type of forces applied to workpieces,
2. Based on the type of products obtained,
3. Based on the temperature of working.

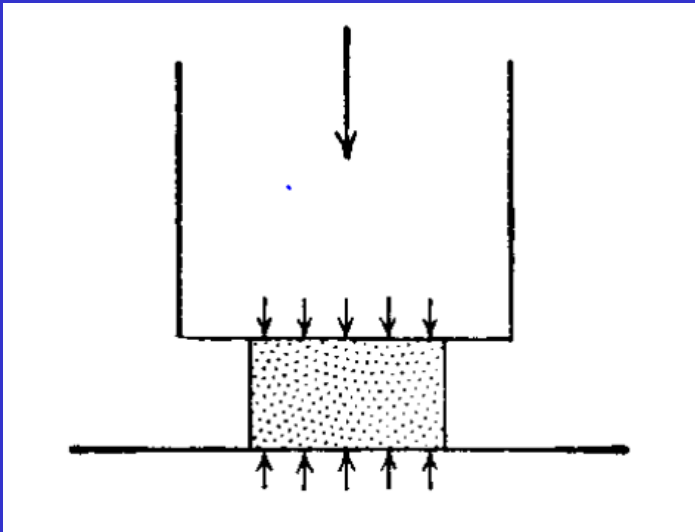
1. Based on the type of forces applied to workpieces,

- a) Direct Compression Type Processes, Ex: - Forging, Rolling
- b) Indirect Compression Type Processes, Ex: - Wire, Tube, Deep Drawing and Extrusion
- c) Tension Type Processes, Ex: - Stretch Forming
- d) Bending Type Processes
- e) Shearing Type Processes

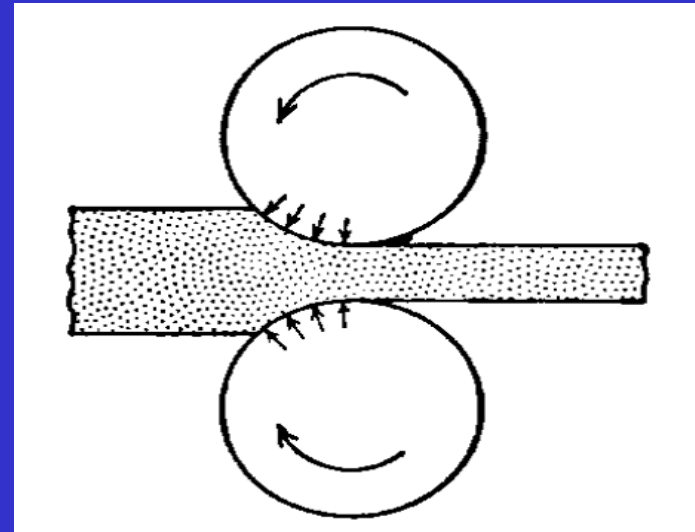
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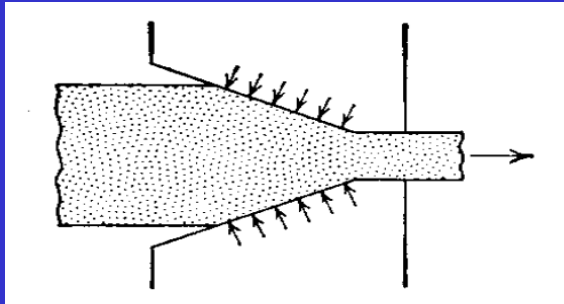
Forging



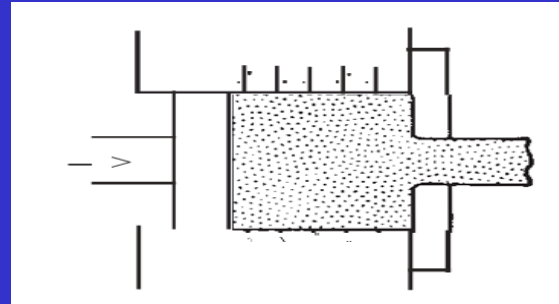
Rolling

Metal flows at right angles to the direction of the compressive force.

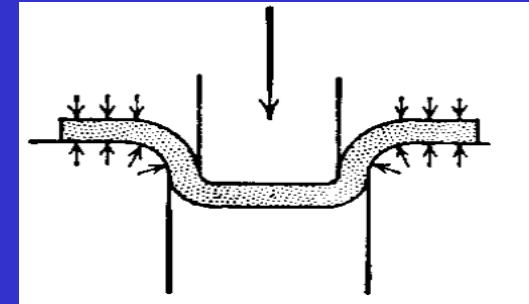
b) Indirect Compression Type Processes, Ex: - Wire, Tube, Deep Drawing and Extrusion



Wire drawing



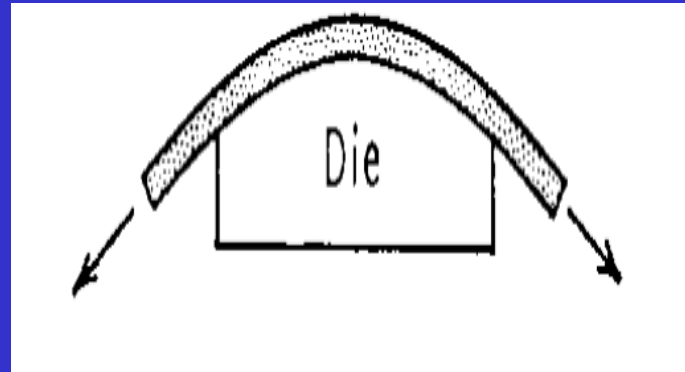
Extrusion



Deep drawing

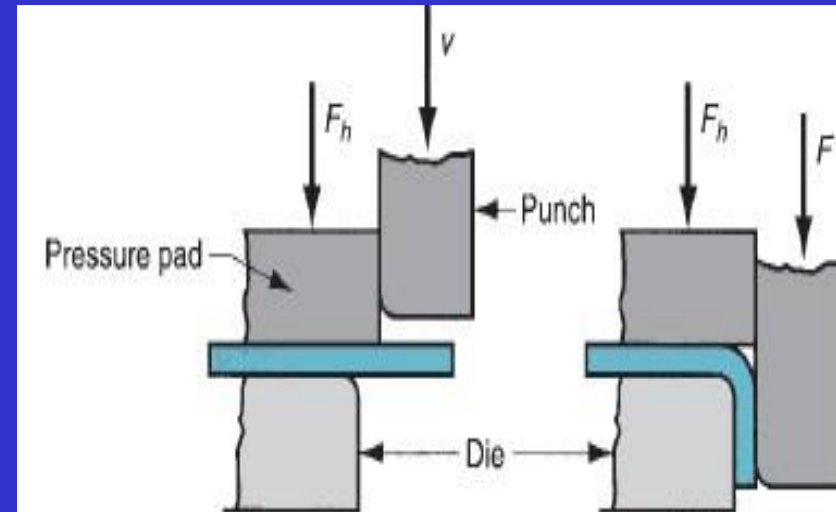
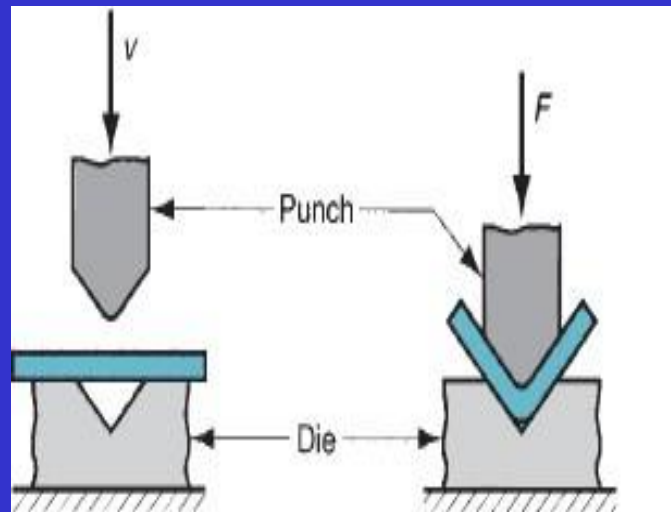
Primary applied forces are frequently tensile (or sometimes compressive), but the indirect compressive forces developed (in at least one of the principal directions) by the reaction of the workpiece with the die aids the deformation process or metal flow.

c) Tension Type Processes, Ex: - Stretch Forming



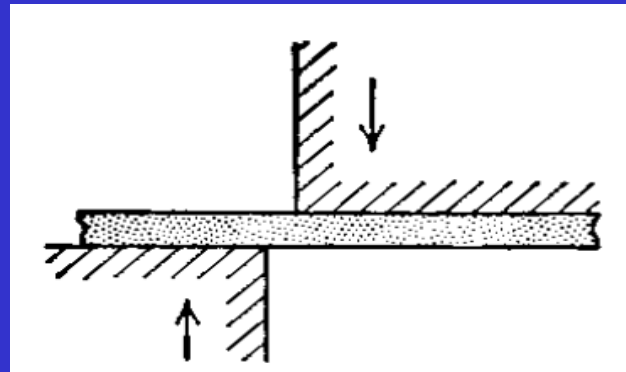
Stretch forming

d) Bending Type Processes



Bending

e) Shearing Type Processes



Shearing

2. Based on the type of products obtained:

a) Primary Mech. Working Processes:- [Processing Operations]

Done in order to get a plate, sheet, billet or ingot, semi - finished or mill products. This will be converted into final shape by some other processes. **Ex:- Rolling , Forging**

b) Secondary Mech. Working Processes:- [Fabrication Operations]

Done in order to get the final product such as wires, components etc...., **Ex:-Wire and Deep Drawing**

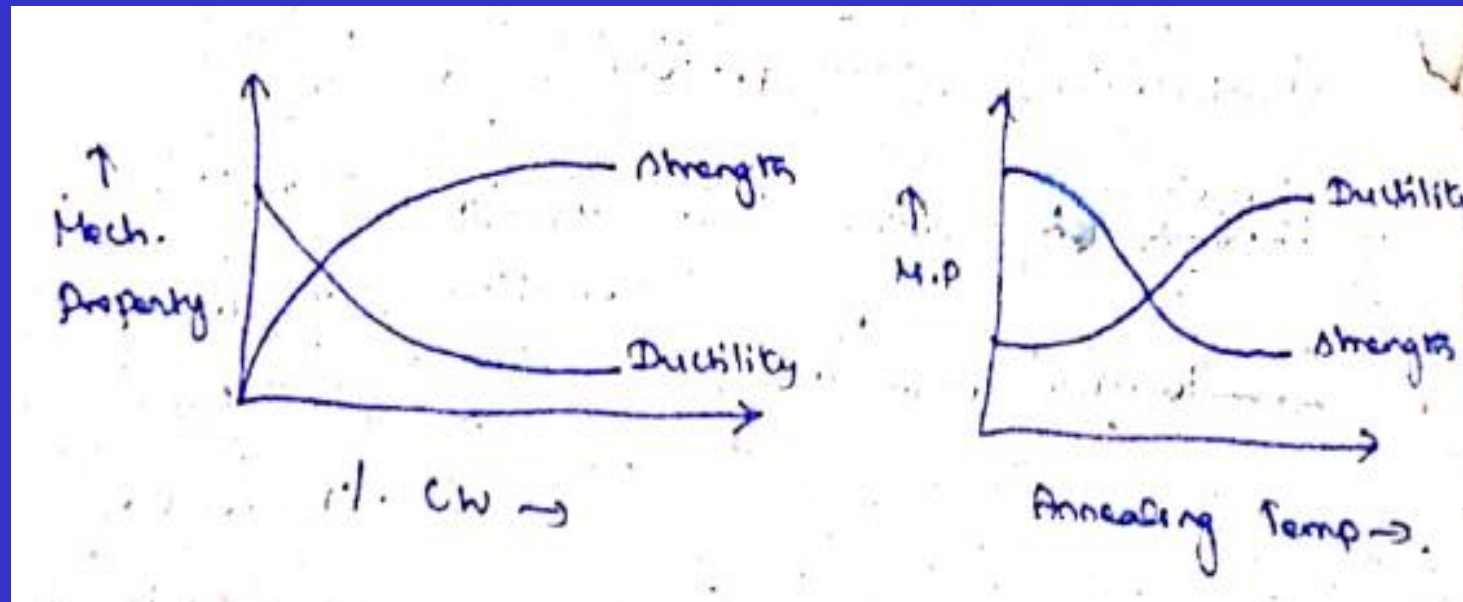


- ❖ Plastic working processes which are designed to reduce an ingot or billet to a standard mill product of simple shape, such as sheet, plate, and bar, are called *primary mechanical working processes*.
- ❖ Forming methods which produce a part to a final finished shape are called *secondary mechanical working processes*.
- ❖ Most sheet-metal forming operations, wire drawing, and tube drawing are secondary processes.
- ❖ Frequently the first category is referred to as *processing operations*, and the second is called *fabrication*.

3. Based on the temperature of working:

a) Cold Working - RT to $< 0.2 T_m$ in K

1. Strain Hardening is possible
2. Dislocation density increases
3. Annealing is a must for further reduction



Advantages:-

1. Excellent surface finish
2. Manipulation of mechanical properties
3. Good dimensional accuracy & Tolerance
4. Since deformation is at low temperature → No scaling or Oxidation → Less wastage of materials
5. Automation is possible
6. Thin gauge sheets can be easily produced
7. Strength can be controlled by appreciable amount of deformation.

Disadvantages:-

1. Require large force to deform, so that sophisticated equipments are required, so cost increases.
2. Several interstage Annealing
3. Stress relieving process is required

b) Hot Working - $> 0.5 T_m$ in K

- ❖ Plastically deforming the material above the recrystallization temperature, so that strain hardening is relieved.
- ❖ Deformation occurs at constant flow stress because strain hardening is relieved.

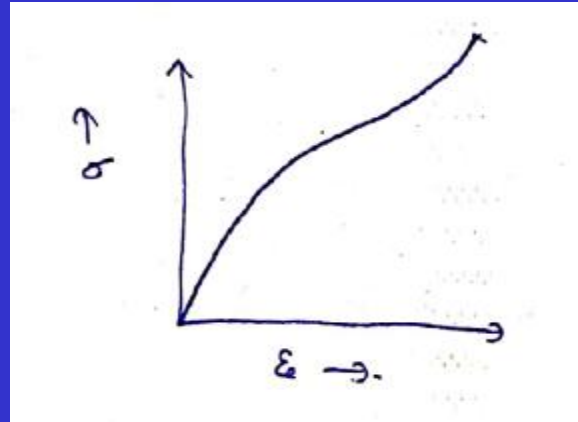
Advantages:-

- ❖ Strain involved is high ($\varepsilon = 2 \text{ to } 4$)
- ❖ Strain rate is also high ($0.5 - 500 \text{ s}^{-1}$)
- ❖ Coarse columnar grains are broken out and equiaxed grains are produced
- ❖ Blow holes are eliminated
- ❖ Segregation and coring (due to difference in melting point) are eliminated
- ❖ Less energy is required to deform the metal
- ❖ Increase in ductility and toughness

Disadvantages:-

- ❖ Scaling is possible , so that oxides are rolled into the material structure
- ❖ High temperature furnaces are required
- ❖ Decarburization is possible [only for steel]
- ❖ Dimensional tolerance is not possible
- ❖ Structure and properties are not uniform

Flow Curve: - [True stress - True strain curve]



- This curve indicates the actual deformation of a metal.
- Flow stress is nothing but the stress required for a particular amount of deformation

Forming load is the load required to produce a particular object, depends upon

1. Geometry of the object
2. Frictional condition
3. Flow stress of a metal

Flow stress depends upon

1. Temperature
2. strain
3. strain rate
4. Metallurgical factors such as chemical composition, grain size, Microstructure segregation etc...,

Yield Criteria or Yield Theory:-

Purpose:-

To predict the onset [i.e., start] of yielding under complex state of stresses [i.e., $\sigma_1, \sigma_2, \sigma_3$]

For ductile metals, there are 2 common theories regarding plastic yielding.

1. Maximum shear stress theory or Tresca Criterion
2. Distortion Energy or von – Mises' Criterion

Tresca Criterion:-

This criterion postulates that yielding will occur when the maximum shear stress reaches a critical value.
so, we assume that $\sigma_1 > \sigma_2 > \sigma_3$.

This criterion predicts yielding when, $\frac{\sigma_1 - \sigma_3}{2} = \tau_{max}$.

$$\therefore \sigma_1 - \sigma_3 = c$$

To evaluate C, a state of uniaxial tension may be used.

In uniaxial tension, $\sigma_{max} = \sigma_1$; $\sigma_2 = \sigma_3 = 0$

Yielding occurs when $\sigma_1 = y$ (or) σ_0 , y (or) $\sigma_0 \rightarrow$ Yield stress of a metal.

$\rightarrow$ Thus, $\sigma_1 - \sigma_3 = y - 0 = c$.

Y = yield stress in simple tension

K = yield stress in pure shear

In case of pure shear, $\sigma_{max} = \sigma_1$

$$\sigma_{min} = \sigma_3 = -\sigma_1$$

$$\sigma_2 = 0.$$

Yielding occurs when maximum shear stress reaches the yield strength in pure shear, i.e. shear yield strength k . → At that time $\sigma_1 = k$. So,

$$\sigma_1 - \sigma_3 = 2k \{ \because \sigma_3 = -\sigma_1 \}$$

Thus, Tresca criterion becomes

$$\sigma_1 - \sigma_3 = y = 2k$$

Thus, yield stress in pure shear (k) is one-half the yield stress in pure tension (y).

1. A material is tested under the state of stress $\sigma_1 = 3\sigma_2 = -2\sigma_3$. Yielding is observed at $\sigma_2 = 140$ MPa. (a) What is the yield stress in simple tension and simple shear? Identify whether yielding takes place in the material? (b) If the material is used under conditions such that $\sigma_1 = -\sigma_3$, $\sigma_2 = 0$, at what value of σ_3 will yielding occur?

von – Mises' Criterion:-

This theory was proposed by Huber in 1904 and was also independently developed by Von- Mises by mathematical terms and interrupted by Henry in 1925.

According to this theory, yielding occurs when the distortion strain energy equals to that associated with yielding in simple tension.

Distortion strain energy = Total strain energy – Volumetric strain energy

$$\text{strain energy} = \frac{1}{2} * \text{stress} * \text{strain}$$

Considering a 3-dimensional stress system of $\sigma_1 > \sigma_2 > \sigma_3$.

$$\text{volumetric strain energy}(U_v) = \frac{1}{2} * \text{mean stress} * \text{volumetric strain}$$

$$\text{volumetric strain}(e_v) = e_1 + e_2 + e_3$$

$$\left[\frac{\sigma_1}{E} - \frac{\nu(\sigma_2 + \sigma_3)}{E} \right] + \left[\frac{\sigma_2}{E} - \frac{\nu(\sigma_3 + \sigma_1)}{E} \right] + \left[\frac{\sigma_3}{E} - \frac{\nu(\sigma_2 + \sigma_2)}{E} \right] \quad \dots (2)$$

$$= \frac{\sigma_1 + \sigma_2 + \sigma_3}{E} (1 - 2\nu)$$

$$\therefore U_v = \frac{1}{2} * \left[\frac{\sigma_1 + \sigma_2 + \sigma_3}{3} \right] \left[\frac{\sigma_1 + \sigma_2 + \sigma_3}{E} (1 - 2\nu) \right]$$

$$U_v = \frac{(\sigma_1 + \sigma_2 + \sigma_3)^2}{6E} (1 - 2\nu)$$

$$\text{Total strain energy} = \frac{1}{2} * \sigma_1 e_1 + \frac{1}{2} * \sigma_2 e_2 + \frac{1}{2} * \sigma_3 e_3$$

Substitute e_1, e_2, e_3 from equation (2), we get

$$U = \frac{1}{2E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)]$$

We know that,

Distortion strain energy = Total strain energy – Volume strain energy

$$U_d = \frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) - \frac{(\sigma_1 + \sigma_2 + \sigma_3)^2}{6E} (1 - 2\nu) \right]$$

We know that, $E = 2G(1 + \nu)$; G = Modulus of rigidity

$$U_d = \frac{1}{12G} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

Or we can call, $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = \text{constant}$

Now, for simple tension, $\sigma_1 = y; \sigma_2 = \sigma_3 = 0$

For simple torsion and pure shear, $\sigma_1 = -\sigma_3 = k, \sigma_2 = 0$

So that von Mises' criteria becomes

$$\begin{aligned} (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 &= 2y^2 (\text{for tension}) \\ &= 6k^2 (\text{for pure shear}) \end{aligned}$$

Thus Von- Mises criterion predicts that yielding starts when $k = \frac{y}{\sqrt{3}} [\because 2y^2 = 6k^2]$

i.e. yield stress in pure shear is $\frac{1}{\sqrt{3}}$ of yield stress in simple tension.

This is 15% higher than predicted by Tresca criterion.

Von- Mises criterion finds considerable experiments results easy to apply than Tresca criterion, since it is based on stress differences only.

However, if the relative magnitude of principal stresses are known, then Tresca criterion is easy to apply.

BY Tresca criterion, $k = \frac{y}{2}$

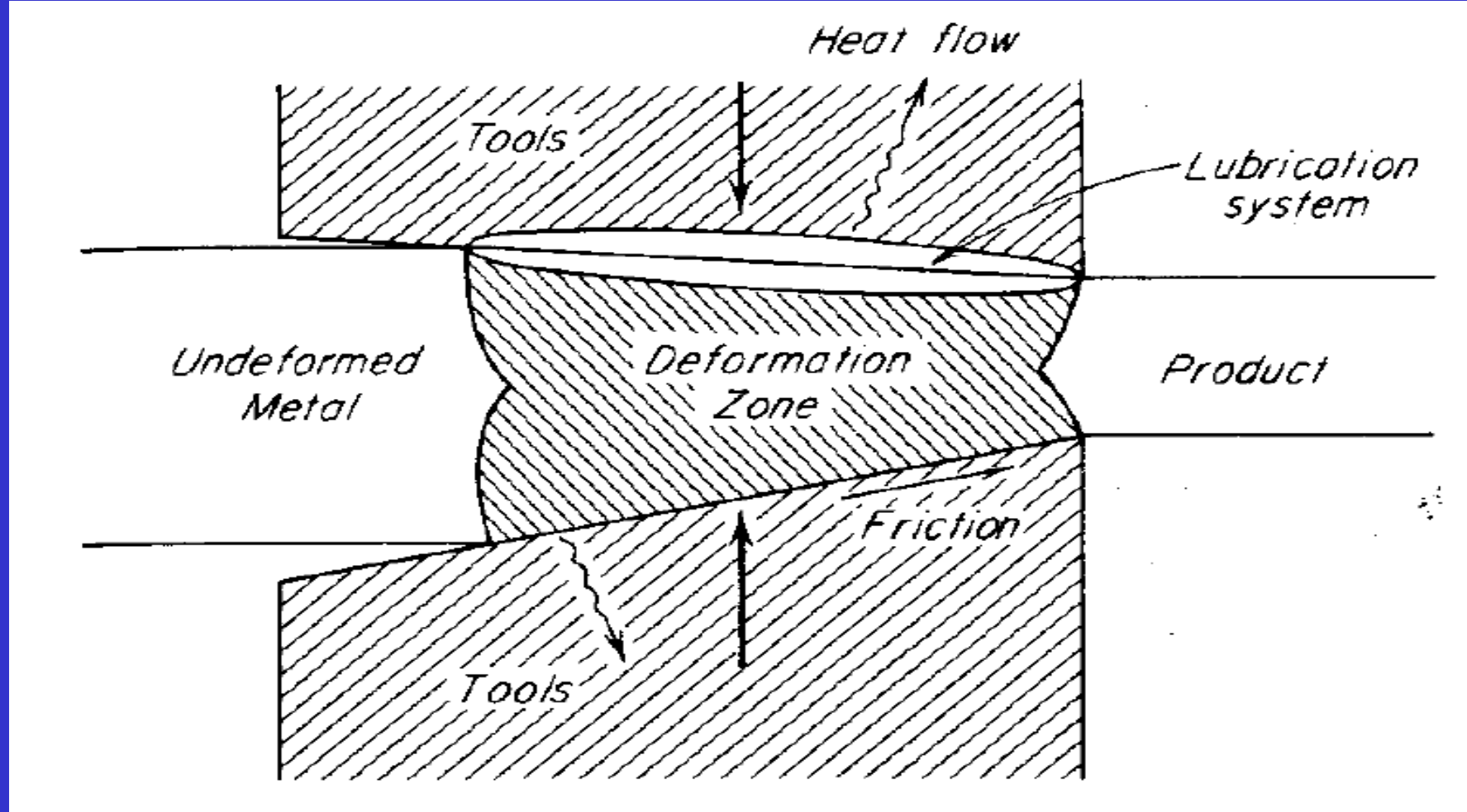
By von Mises' criterion, $k = \frac{y}{\sqrt{3}}$

If the yield strength of a steel is 950 MPa, determine whether yielding will have occurred on the basis of both the von-Mises' and Tresca criteria. The stress state is given by

$$\begin{bmatrix} 0 & 0 & 300 \\ 0 & -400 & 0 \\ 300 & 0 & -800 \end{bmatrix} \text{ MPa.}$$

Deformation Process:-

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Deformation processing system .

- ❖ The deformation zone is concerned with the distribution of stress, strain, and particle velocities, and with the overall pressure required to perform the operation.
- ❖ Workpiece will be in contact with nondeforming (elastic) tools or dies. The friction along the interface and the heat transfer from the workpiece to the die are important considerations, as they affect the tool wear and surface finish of the product.
- ❖ The applied forces must develop yielding in the material, but the stresses must not locally create fracture.
- ❖ The flow stress of the material is a strong function of strain, strain rate, and temperature.
- ❖ Metallurgical phenomena such as strain hardening, recrystallization and fracture are predominant, but often under specialized conditions of high strain rates and/ or high temperature.

Effect of variables on Metal Forming:-

1. Temperature
2. Strain rate
3. Microstructure
4. Effect of friction

Recovery and Recrystallization:-

Dynamic: The recovery and recrystallization occurs along with the deformation

Static: The recovery and recrystallization occurs without the deformation process for a gradual increase in temperature.

Ex:- Heating a metal slowly in a furnace.

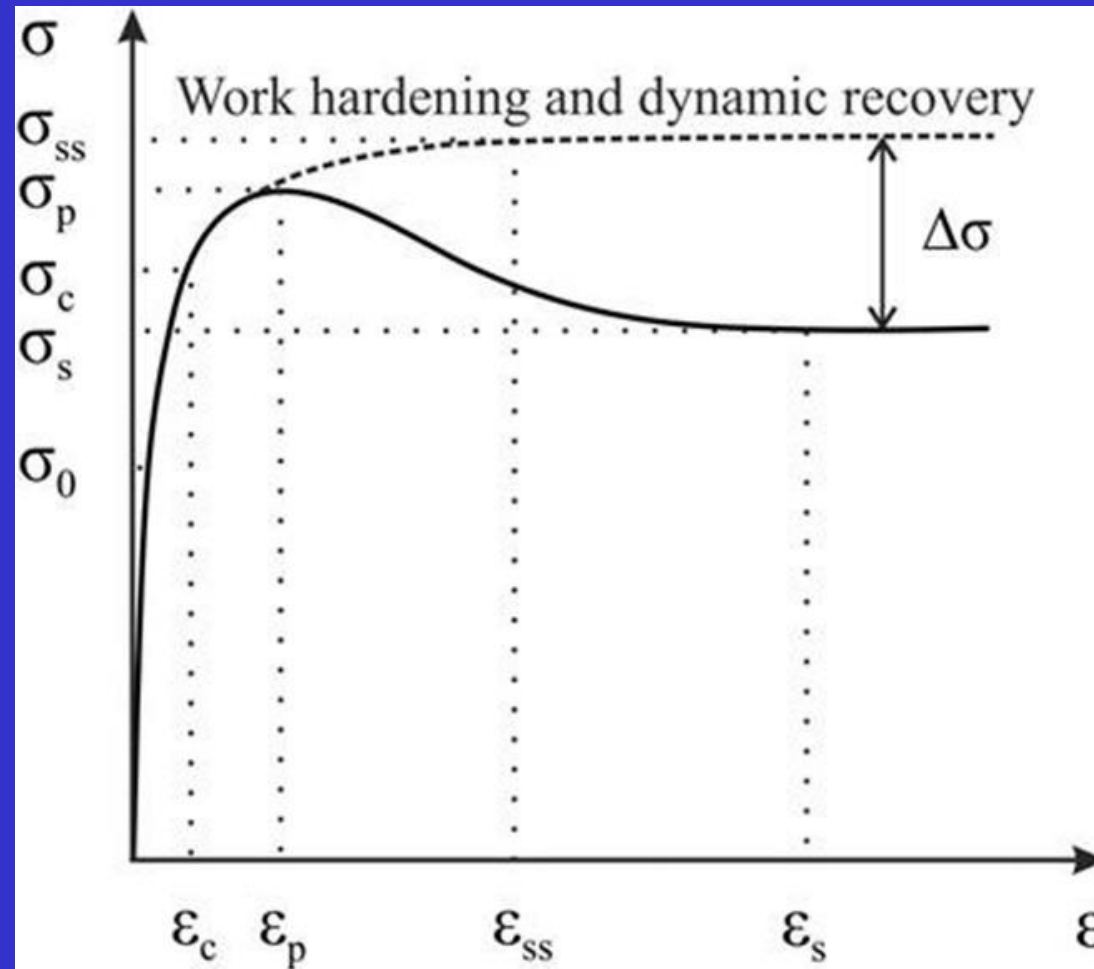
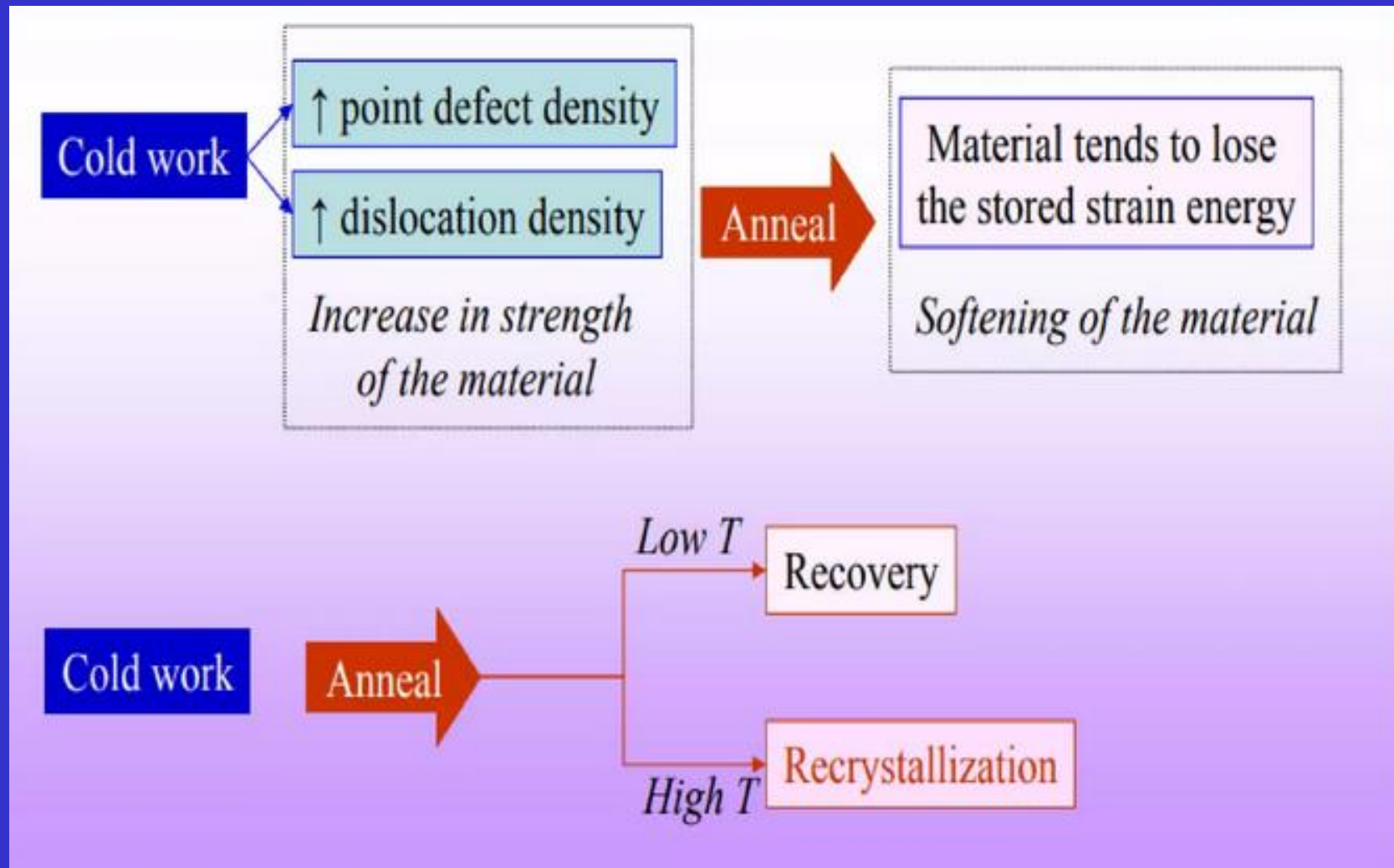
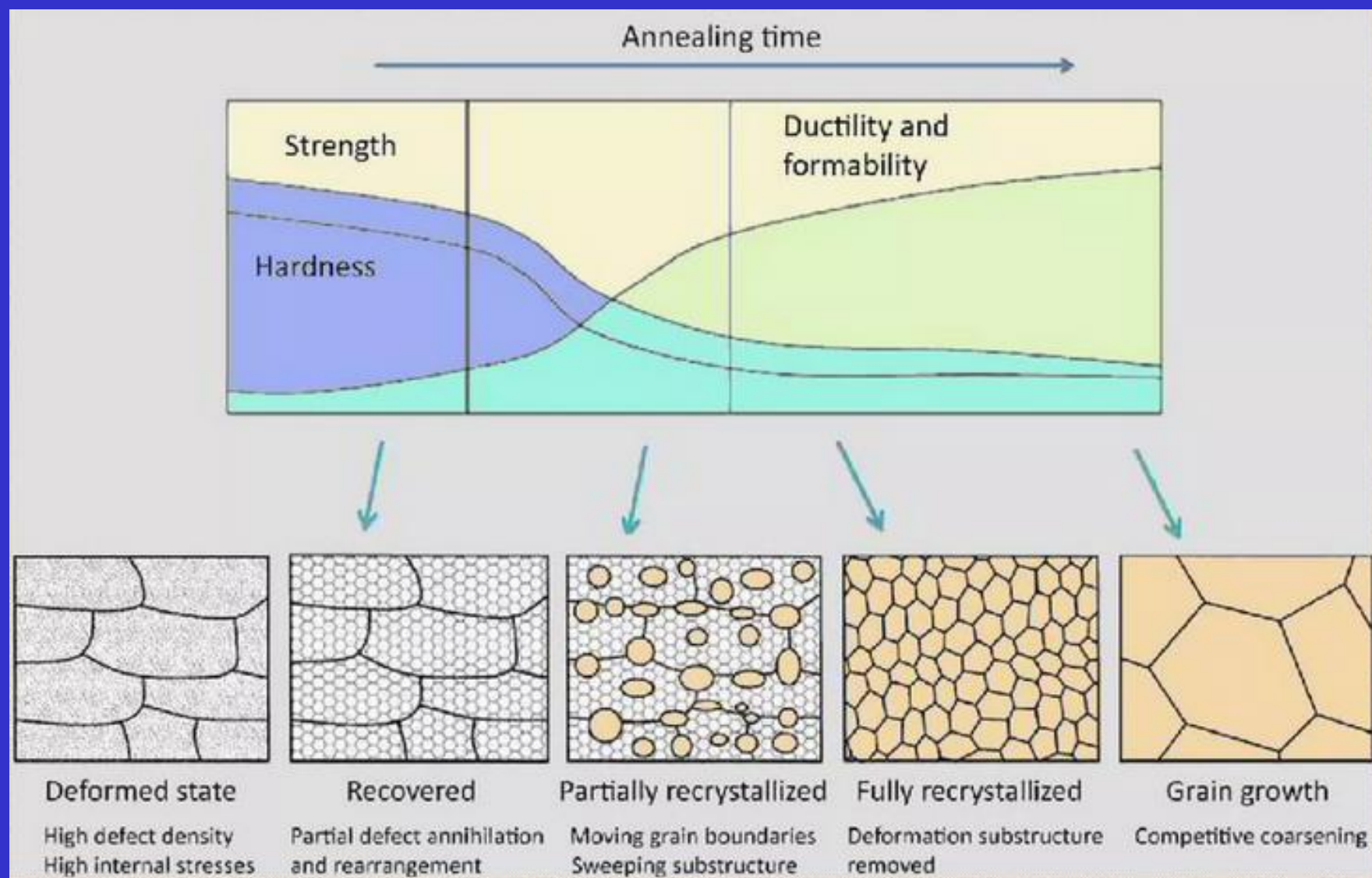


Illustration of the flow curves for the work hardening, dynamic recovery (dashed line), and softening due to dynamic recrystallization (solid line).





The temperature of a metal workpiece during metal working depends on,

1. Initial temperature of the tool and material.
2. Heat generated due to plastic deformation.
3. Heat generated by friction between die and material interface.
4. Heat transfer between the deforming material, dies and surrounding material.

Final average temperature of metal after time t , $T_m = T_d + T_f + T$

$T_d \rightarrow$ Increase in temperature due to plastic deformation

$T_f \rightarrow$ Increase in temperature due to friction

$T \rightarrow$ Average instantaneous temperature of deforming metal at interface

- i. For frictionless deformation process, the maximum increase in temperature is

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$$T_d = \frac{U_p}{\rho c J} = \frac{\bar{\sigma} \bar{\epsilon} \beta}{\rho c J}$$

U_p = Work done due to plastic deformation/ unit volume

ρ = Density of workpiece

c = Specific heat of workpiece

β = Fraction of deformation work converted into heat, roughly 0.95
The remainder is stored in the material as energy associated with the defect structure.

J = Mechanical equivalent of heat, 4185 J/kcal.

ii. The temperature increase due to friction is given by

$$T_f = \frac{\mu \cdot P \cdot v \cdot A \cdot \Delta t}{\rho c V J}$$

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μ = Friction co-efficient at metal – tool interface

P = Stress normal to interface

v = Velocity at metal –tool interface.

A = Surface area at metal – tool interface

Δt = Time interval of consideration

V = Volume of metal subjected to temperature rise

iii. Average instantaneous temperature of deforming metal at interface

$$T = T_1 + (T_0 - T_1) \cdot \exp\left(\frac{-ht}{\rho c \delta}\right)$$

$T_0 \rightarrow$ Initial workpiece temperature; $T_1 \rightarrow$ Die temperature

$H \rightarrow$ heat transfer coefficient between die & material; $\delta \rightarrow$ material thickness between the dies.

Example Compare the temperature rise when a cylinder of aluminum and titanium is quickly deformed to $\bar{\epsilon} = 1.0$ at room temperature.

	$\bar{\sigma}$, MPa	$\bar{\epsilon}$	ρ , kg m ⁻³	c , J kg ⁻¹ K ⁻¹
Al	200	1.0	2690	900
Ti	400	1.0	4500	519

Strain Rate Effects:-

$$\dot{\varepsilon} = \frac{d\varepsilon}{dt} / sec$$

Strain Rate or Deformation Velocity has 3 effects on manufacturing processes:-

1. Flow stress of a material increases with Strain Rate [Strain Rate increases, Flow Stress increases]
2. Temperature of a metal increases because of adiabatic heating of metal.
3. There is improved lubrication at tool – metal interface as long as lubricant film is maintained.

Definition of Strain Rate: -

1. True Strain Rate for a cylinder upset in compression is given by

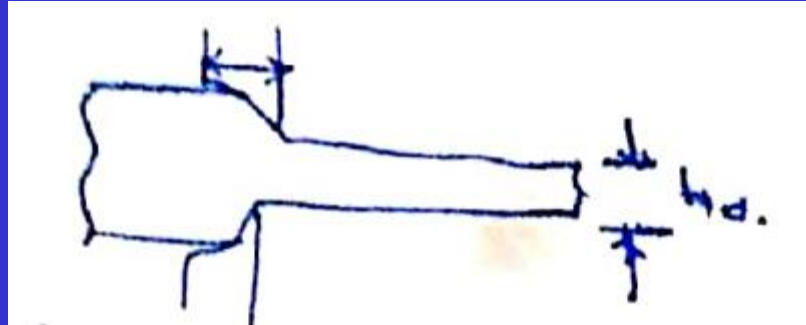
$$\dot{\varepsilon} = \frac{d\varepsilon}{dt} = \frac{d h/h}{dt} = \frac{1}{h} \cdot \frac{dh}{dt} = \frac{v}{h}$$

v = Deformation velocity

$$\therefore \dot{\varepsilon} = \frac{\text{Deformation velocity}}{\text{Instantaneous height}}$$

2. Mean Strain Rate:-

A workpiece is deformed by converging dies where the height varies with distance between the dies



$$\dot{\bar{\epsilon}}_x = \frac{1}{L} \int_0^L \dot{\epsilon} dx$$

L = length of contact between tool and workpiece

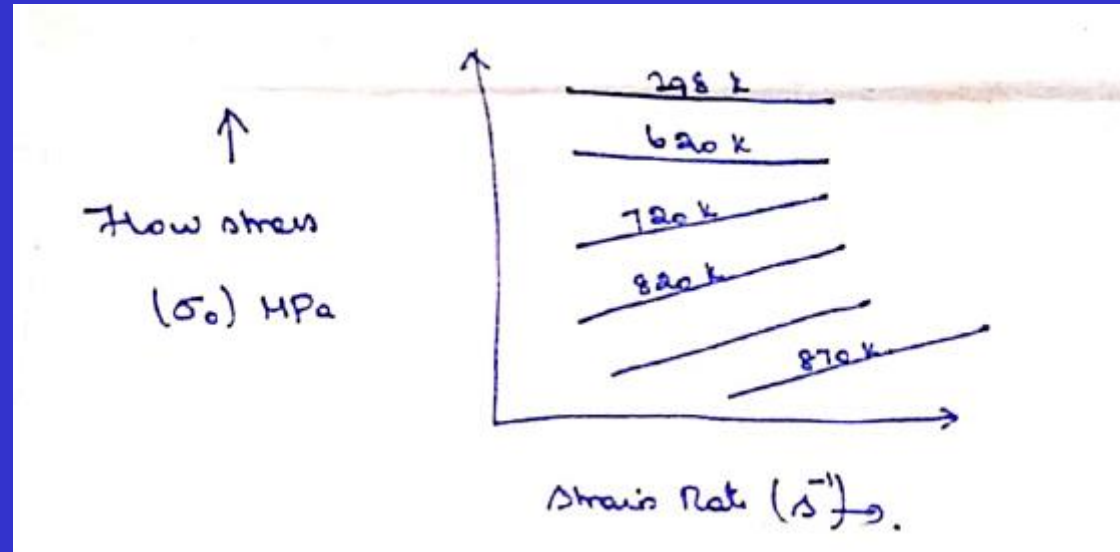
3. Mean Strain Rate in terms of time for an element to travel through the die

$$\dot{\bar{\varepsilon}}_t = \frac{1}{t_1} \int_0^{t_1} \dot{\varepsilon} dt$$

4. Root Mean Power Strain Rate [for Strain Rate sensitive material of large reduction]

$$\dot{\varepsilon}_{rmp} = \left[\frac{1}{\ln R} \int_0^{\ln R} (\dot{\varepsilon})^n \cdot d\varepsilon \right]^{1/n}$$

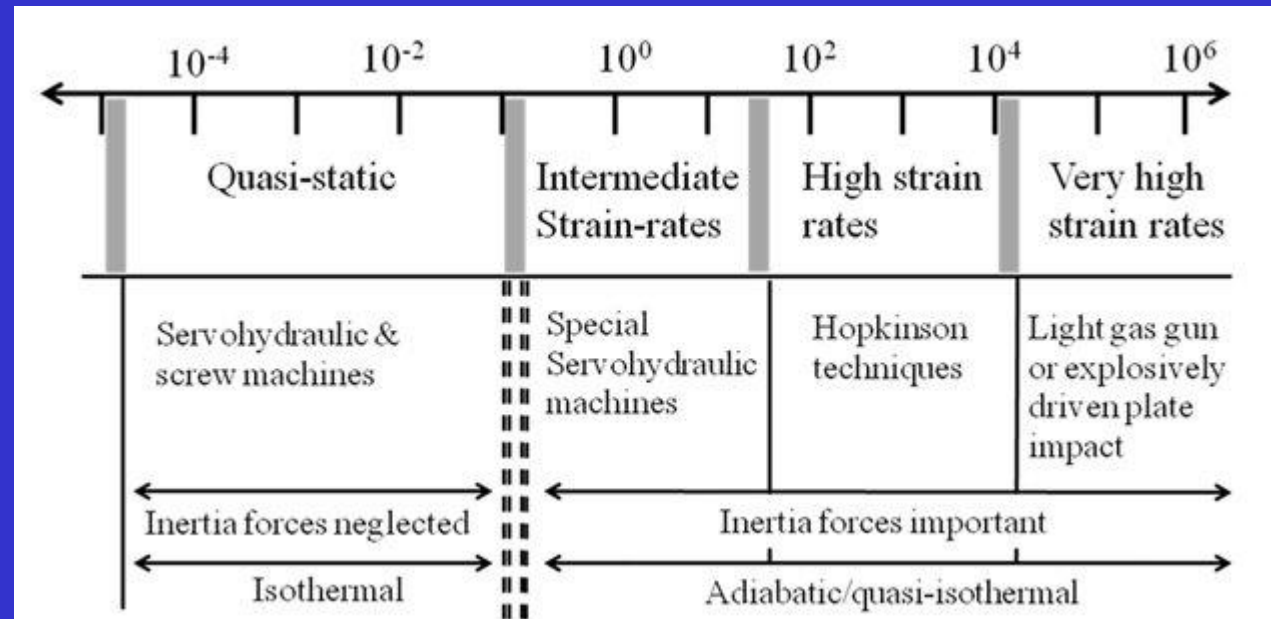
Where $R = \frac{A_0}{A}$ = deformation ratio
n = Strain rate sensitivity



Strain Rate has a greater influence on the yield point of a specimen. In case of mild steel tension test, by applying a slow Strain Rate, we will not get yield point.

Strain Rate $\propto$ Strength

Operation	Velocity, ft/s
Tension Test	2×10^{-6} to 2×10^{-2}
Hydraulic extrusion press	0.01 to 10
Mechanical press	0.5 to 5
Charpy impact test	10 to 20
Forging hammer	10 to 30
Explosive	100 to 400



Relationship between σ and $\dot{\epsilon}$ [Flow stress and Strain Rate]

$$\sigma = c(\dot{\epsilon})^n |_{\epsilon, t}$$

Where C = constant
 n = Strain rate sensitivity

$$\sigma = k_1 + k_2 \ln \frac{\dot{\epsilon}}{\dot{\epsilon}_0}$$

$k_1, k_2, \dot{\epsilon}_0$ are constants

Strain Rate sensitivity is a good indicator of changes in deformation behavior.

Cross – head velocity is

$$U = \frac{dL}{dt}$$

Conventional Linear Strain Rate: -

$$\dot{e} = \frac{de}{dt} = \frac{d\left(\frac{L - L_0}{L_0}\right)}{dt} = \frac{1}{L_0} \cdot \frac{dL}{dt} = \frac{u}{L_0}$$

True Linear Strain Rate:-

$$\dot{\varepsilon} = \frac{d\varepsilon}{dt} = \frac{d\left(\ln \frac{L}{L_0}\right)}{dt} = \frac{1}{L_0} \cdot \frac{dL}{dt} = \frac{u}{L}$$

For Constant v ,

If L is increased, then $\dot{\epsilon}$ will decrease. But in order to maintain it constantly, v should not be kept constant. its value must also be increased.

Devices used for object constant Strain Rate: -

1. Open loop control - up to uniform plastic deformation region i.e. below UTS.
2. Closed loop control - above uniform i.e. localized deformation region i.e. above UTS.

HERF:- [High Energy Rate Forming Process] – Strain Rate is high

By applying high energy to the material is workpiece, High Strain Rate is obtained. This energy may be by explosion of materials. Ex:- Explosive forming type

Super Plasticity: -

Higher amount of plastic deformation i.e. elongation may be from 100-1000%

Low flow stress is required.

Conditions to obtain Super plasticity: -

1. Fine grain size, $G.S < 1\mu m$
2. Very high temperature, $Temp > 0.4T_m$
3. Low strain rate, $Strain\ rate < 0.1\ s^{-1}$

It is used for metals which is difficult to work.

The methods of analysis, in increasing order of complexity and ability to predict fine detail, are:

1. The **slab method** assumes homogeneous deformation
2. **Uniform-deformation energy method** calculates average forming stress from the work of plastic deformation
3. **Slip-line field theory** permits point-by-point calculation of stress for plane strain conditions only
4. **Upper and lower-bound solutions** based on the theory of limit analysis, uses reasonable stress and velocity fields to calculate the bounds within which the actual forming load must lie
5. **Finite element methods** a technique called the matrix method allows large increments of deformation for rigid plastic materials, with considerable reduction in computational time

Effect of Friction and Lubrication:-

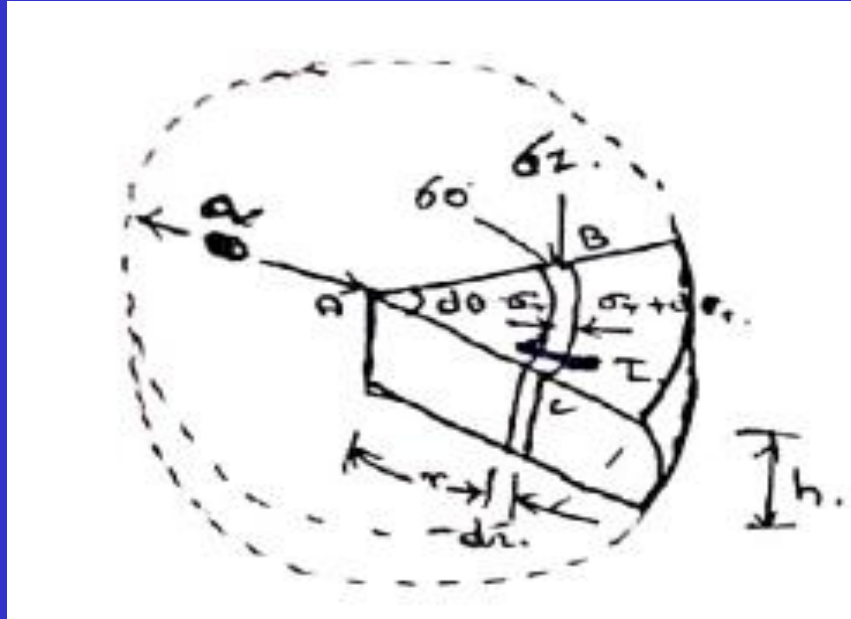
$$\text{Co-efficient of friction } (\mu) = \frac{\tau}{P} = \frac{\text{shear stress at the interface}}{\text{stress normal to the interface}}$$



Coulomb's law of friction

Slab Analysis of Homogeneous Compression of a flat circular disk :- [with friction]

Consider a small radial elemental volume at a distance “ r ” from the centre and making an angle “θ”



Total radius of disk “a”

Height of disk “h”

Assumptions: -

1. No value of barreling of the edges of the disk

2. Thickness is small enough that the axial compressive stress σ_z is constant throughout the thickness.

Stresses acting on this element are: -

1. Compressive stress (from the forging press) or Vertical stress = σ_z
2. Stress acting on the element in Radial direction = σ_r & $\sigma_r + d\sigma_r$
3. Circumferential stress = σ_θ
4. Frictional shear stress acting radially inward and acting on top and bottom surface.

Referring to the figure and taking equilibrium of forces in the radial direction, i. e. $\sigma_z = 0$

Equilibrium of forces means all the forces acting on the material should be equal to zero.

Area of radial element = Arc area * Radial thick/gap (dr)

$$\sigma_r \cdot (rd\theta \cdot h) \downarrow (\sigma_r + d\sigma_r)[(r + dr)d\theta \cdot h] + 2\sigma_\theta \cdot \sin \frac{d\theta}{2} \cdot hdr \downarrow 2\tau(rd\theta dr) = 0$$

Inward direction to compression
Towards inward direction

area of radial element

σ_z is a vertical force and its component in other direction is zero.

So, by simplifying we get

$$\sigma_r \cdot (rd\theta \cdot h) - (\sigma_r + d\sigma_r)[r \cdot d\theta \cdot h + dr \cdot d\theta \cdot h] + 2\sigma_\theta \cdot \sin \frac{d\theta}{2} \cdot hdr - 2\tau(rd\theta dr) = 0$$

$$-\sigma_r \cdot dr \cdot d\theta \cdot h - d\sigma_r \cdot r \cdot d\theta \cdot h - d\sigma_r \cdot dr \cdot d\theta \cdot h + 2\sigma_\theta \cdot \sin \frac{d\theta}{2} \cdot hdr - 2\tau(rd\theta dr) = 0$$

$$\sin \frac{d\theta}{2} = \frac{d\theta}{2} \{for \text{ small values of } d\theta\}$$

Dividing throughout by $-d\theta$

$$h.\sigma_r.dr + d\sigma_r.r.h + d\sigma_r.dr.h - \sigma_\theta.h.dr + 2\tau(r.dr) = 0$$

It can be seen that from axial symmetry of Homogenization strain is circumferential direction = strain in radial direction.

$$d\varepsilon_\theta = d\varepsilon_r \text{ and from plasticity theory } \sigma_\theta = \sigma_r$$

Making the substitute, we arrive at

$$\sigma_r.dr.h - \sigma_r.dr.h + d\sigma_r.r.h + d\sigma_r.dr.h + 2\tau(r.dr) = 0$$

$$d\sigma_r.r.h + d\sigma_r.dr.h + 2\tau(r.dr) = 0$$

Dividing by $h.r.dr$

$$\frac{d\sigma_r}{dr} + \frac{2\tau}{h} = 0 \quad \dots(1)$$

[Double integration term is neglected]

(iii) From the coulomb's friction law

$$\mu = \frac{\tau}{P}$$

$$\tau = \mu P$$

$$[\because P = \sigma_z]$$

Substitute in (1), we get,

$$\frac{d\sigma_r}{dr} + \frac{2\mu\sigma_z}{h} = 0 \quad \dots (2)$$

By making the assumptions that $\sigma_r, \sigma_\theta, \sigma_z$ as $\sigma_1, \sigma_2, \sigma_3$, respectively, are principle stresses, we can utilize the von-Mises' Criteria, and develop a relationship between σ_r & σ_z

Noting that σ_r & σ_θ are same , we can write

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2y^2 \text{ or } 2\sigma_0^2$$

$$(\sigma_r - \sigma_\theta)^2 + (\sigma_\theta - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 = 2\sigma_0^2$$

$$\sigma_r = \sigma_\theta \{ \text{To eliminate } \sigma_\theta \}$$

$$\sigma_r^2 + \sigma_z^2 - 2\sigma_r\sigma_z + \sigma_r^2 + \sigma_z^2 - 2\sigma_r\sigma_z = 2\sigma_0^2$$

$$2(\sigma_r^2 + \sigma_z^2 - 2\sigma_r\sigma_z) = 2\sigma_0^2$$

$$(\sigma_r - \sigma_z)^2 = \sigma_0^2$$

$$\sigma_r - \sigma_z = \sigma_0 \quad \dots (3)$$

Differentiate

$$d\sigma_r = -dp \quad \dots (4)$$

Substitute (4) in (2)

$$\frac{-dp}{dr} + \frac{2\mu(-P)}{h} = 0$$

$$\frac{dp}{dr} + \frac{2\mu(P)}{h} = 0$$

$$\frac{dp}{dr} = -\frac{2\mu(P)}{h}$$

$$\frac{dp}{p} = -\frac{2\mu}{h} dr$$

Integrating, we get

$$\ln P = -\frac{2\mu}{h} r + C \quad \dots(4a)$$

To find the value of C , we want to apply the boundary conditions

B.C: *At the outer surface of the disk, $r = a, \sigma_r = 0$*

Applying in equation 3a, $P = \sigma_0$

Applying both B.C's in equations

$$\ln \sigma_0 = -\frac{2\mu}{h}a + C$$

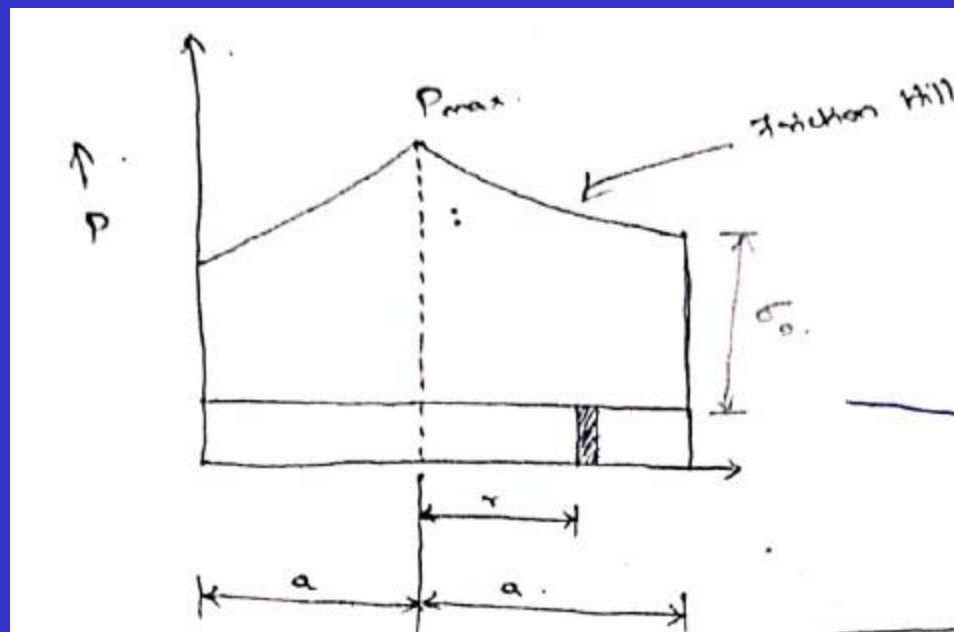
$$C = \ln \sigma_0 + \frac{2\mu a}{h}$$

Substitute in eq (4a)

$$\ln P = -\frac{2\mu}{h}r + \ln \sigma_0 + \frac{2\mu a}{h}$$

$$\ln \left(\frac{P}{\sigma_0} \right) = \frac{2\mu}{h} [a - r] \quad \dots (5)$$

So, the axial compressive stress P is plotted against the diameter of the disc



We get maximum compressive stress at $r = 0$ $P_{max} = \sigma_0 \cdot e^{\frac{2\mu}{h}[a]}$

We get minimum compressive stress at $r = a$, $P_{min} = \sigma_0$

The pressure distribution is symmetrical about the central line and tries to a sharp peak at center of the disk.

This characteristic rise in deformation pressure with distance is often called a friction Hill.

The average deformation pressure is given by,

Consider the area of the elemental volume (ABC)= $r.d\theta.dr$

Load is the whole of the elemental volume (ABC)= $P.(2\pi r.dr)$

[Area of whole circle = $2\pi r.dr$]

∴ Total load on the disk

$$P_T = \int_0^a P \cdot 2\pi r \cdot dr$$

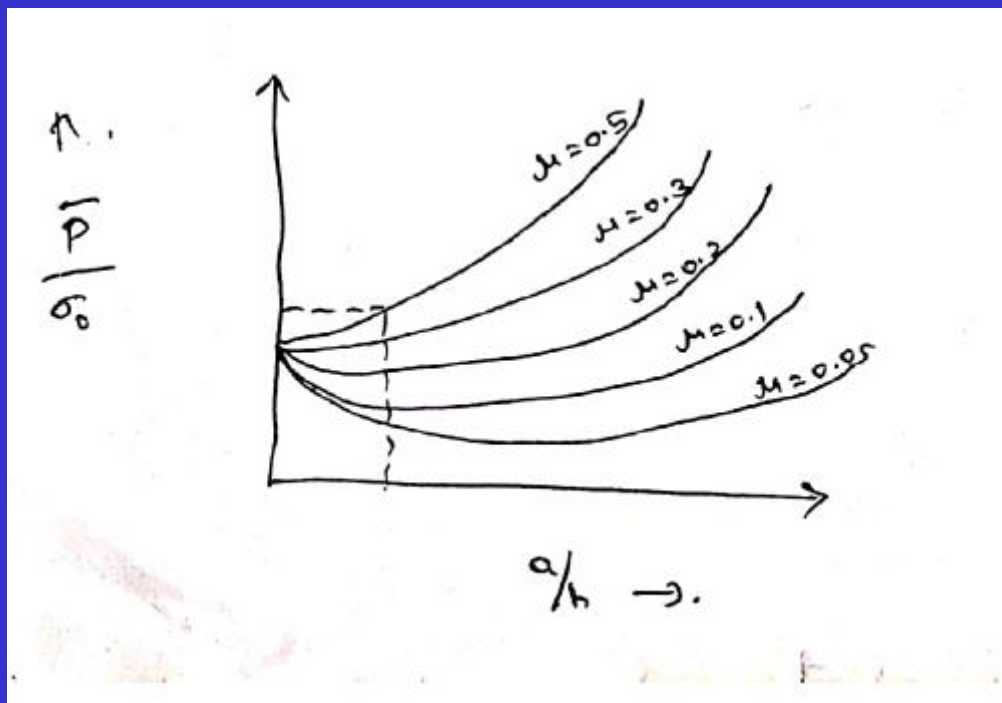
Substitute eq. 6 for P

$$P_T = \frac{\int_0^a P \cdot 2\pi r \cdot dr}{\pi a^2} * \sigma_0 \cdot e^{\frac{2\mu}{h}[a-r]}$$

Average deformation pressure

$$\bar{P} = \frac{\int_0^a P \cdot 2\pi r \cdot dr}{\pi a^2} * \sigma_0 \cdot e^{\frac{2\mu}{h}[a-r]} \frac{\sigma_0}{2} \left(\frac{h}{\mu a} \right)^2 \left[e^{\frac{2\mu}{h}[a]} - \frac{2\mu}{h}[a] - 1 \right] \quad \dots (1)$$

μ = Sliding friction



$\frac{a}{h}$ = dimensionless quantity